

Chapter 2

The DOT and CROSS PRODUCT

THE DOT OR SCALAR PRODUCT of two vectors **A** and **B**, denoted by **A · B** (read **A dot B**), is defined as the product of the magnitudes of **A** and **B** and the cosine of the angle θ between them. In symbols,

$$\underline{\mathbf{A} \cdot \mathbf{B} = AB \cos \theta}, \quad \underline{0 \leq \theta \leq \pi}$$

Note that **A · B** is a scalar and not a vector.

The following laws are valid:

$$1. \underline{\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}} \quad \text{Commutative Law for Dot Products}$$

$$2. \underline{\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}} \quad \text{Distributive Law}$$

$$3. \underline{m(\mathbf{A} \cdot \mathbf{B}) = (m\mathbf{A}) \cdot \mathbf{B} = \mathbf{A} \cdot (m\mathbf{B}) = (\mathbf{A} \cdot \mathbf{B})m}, \quad \text{where } m \text{ is a scalar.}$$

$$4. \underline{\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1, \quad \mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0}$$

$$5. \text{ If } \underline{\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}} \text{ and } \underline{\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}}, \text{ then}$$

$$\underline{\mathbf{A} \cdot \mathbf{B} = A_1B_1 + A_2B_2 + A_3B_3}$$

$$\underline{\mathbf{A} \cdot \mathbf{A} = A^2 = A_1^2 + A_2^2 + A_3^2}$$

$$\underline{\mathbf{B} \cdot \mathbf{B} = B^2 = B_1^2 + B_2^2 + B_3^2}$$

$$6. \text{ If } \underline{\mathbf{A} \cdot \mathbf{B} = 0} \text{ and } \underline{\mathbf{A} \text{ and } \mathbf{B} \text{ are not null vectors}}, \text{ then } \underline{\mathbf{A} \text{ and } \mathbf{B} \text{ are perpendicular.}}$$

THE CROSS OR VECTOR PRODUCT of **A** and **B** is a vector **C** = **A** × **B** (read **A cross B**). The magnitude of **A** × **B** is defined as the product of the magnitudes of **A** and **B** and the sine of the angle θ between them. The direction of the vector **C** = **A** × **B** is perpendicular to the plane of **A** and **B** and such that **A**, **B** and **C** form a right-handed system. In symbols,

$$\underline{\mathbf{A} \times \mathbf{B} = AB \sin \theta \mathbf{u}}, \quad \underline{0 \leq \theta \leq \pi}$$

where **u** is a unit vector indicating the direction of **A** × **B**. If **A** = **B**, or if **A** is parallel to **B**, then $\sin \theta = 0$ and we define **A** × **B** = **0**.

The following laws are valid:

$$1. \underline{\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}} \quad \text{(Commutative Law for Cross Products Fails.)}$$

$$2. \underline{\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}} \quad \text{Distributive Law}$$

$$3. \underline{m(\mathbf{A} \times \mathbf{B}) = (m\mathbf{A}) \times \mathbf{B} = \mathbf{A} \times (m\mathbf{B}) = (\mathbf{A} \times \mathbf{B})m}, \quad \text{where } m \text{ is a scalar.}$$

$$4. \underline{\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}}, \quad \underline{\mathbf{i} \times \mathbf{j} = \mathbf{k}}, \quad \underline{\mathbf{j} \times \mathbf{k} = \mathbf{i}}, \quad \underline{\mathbf{k} \times \mathbf{i} = \mathbf{j}}$$

$$5. \text{ If } \underline{\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}} \text{ and } \underline{\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}}, \text{ then}$$

$$\underline{\mathbf{A} \times \mathbf{B}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

6. The magnitude of $\mathbf{A} \times \mathbf{B}$ is the same as the area of a parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$.

7. If $\mathbf{A} \times \mathbf{B} = \mathbf{0}$, and $\mathbf{A}$ and $\mathbf{B}$ are not null vectors, then $\mathbf{A}$ and $\mathbf{B}$ are parallel.

TRIPLE PRODUCTS. Dot and cross multiplication of three vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ may produce meaningful products of the form $(\mathbf{A} \cdot \mathbf{B})\mathbf{C}$, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ and $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$. The following laws are valid:

1. $(\mathbf{A} \cdot \mathbf{B})\mathbf{C} \neq \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$

2. $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) =$ volume of a parallelepiped having $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ as edges, or the negative of this volume, according as $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ do or do not form a right-handed system. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$ and $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$, then

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}$$

3. $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$

(Associative Law for Cross Products Fails.)

4. $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$
 $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A}$

The product $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ is sometimes called the *scalar triple product* or *box product* and may be denoted by $[\mathbf{ABC}]$. The product $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ is called the *vector triple product*.

In $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ parentheses are sometimes omitted and we write $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$ (see Problem 41). However, parentheses must be used in $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ (see Problems 29 and 47).

RECIPROCAL SETS OF VECTORS. The sets of vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are called *reciprocal sets or systems of vectors* if

$$\mathbf{a} \cdot \mathbf{a}' = \mathbf{b} \cdot \mathbf{b}' = \mathbf{c} \cdot \mathbf{c}' = 1$$

$$\mathbf{a}' \cdot \mathbf{b} = \mathbf{a}' \cdot \mathbf{c} = \mathbf{b}' \cdot \mathbf{a} = \mathbf{b}' \cdot \mathbf{c} = \mathbf{c}' \cdot \mathbf{a} = \mathbf{c}' \cdot \mathbf{b} = 0$$

The sets $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are reciprocal sets of vectors if and only if

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$$

where $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$. See Problems 53 and 54.

SOLVED PROBLEMS

THE DOT OR SCALAR PRODUCT.

1. Prove
- $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- .

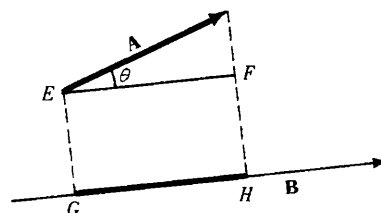
$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta = BA \cos \theta = \mathbf{B} \cdot \mathbf{A}$$

Then the commutative law for dot products is valid.

2. Prove that the projection of
- $\mathbf{A}$
- on
- $\mathbf{B}$
- is equal to
- $\mathbf{A} \cdot \mathbf{b}$
- , where
- $\mathbf{b}$
- is a unit vector in the direction of
- $\mathbf{B}$
- .

Through the initial and terminal points of $\mathbf{A}$ pass planes perpendicular to $\mathbf{B}$ at G and H respectively as in the adjacent figure; then

$$\text{Projection of } \mathbf{A} \text{ on } \mathbf{B} = \overline{GH} = \overline{EF} = A \cos \theta = \mathbf{A} \cdot \mathbf{b}$$



3. Prove
- $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- .

Let $\mathbf{a}$ be a unit vector in the direction of $\mathbf{A}$; then

Projection of $(\mathbf{B} + \mathbf{C})$ on $\mathbf{A} = \text{proj. of } \mathbf{B} \text{ on } \mathbf{A} + \text{proj. of } \mathbf{C} \text{ on } \mathbf{A}$

$$(\mathbf{B} + \mathbf{C}) \cdot \mathbf{a} = \mathbf{B} \cdot \mathbf{a} + \mathbf{C} \cdot \mathbf{a}$$

Multiplying by A ,

$$(\mathbf{B} + \mathbf{C}) \cdot A\mathbf{a} = \mathbf{B} \cdot A\mathbf{a} + \mathbf{C} \cdot A\mathbf{a}$$

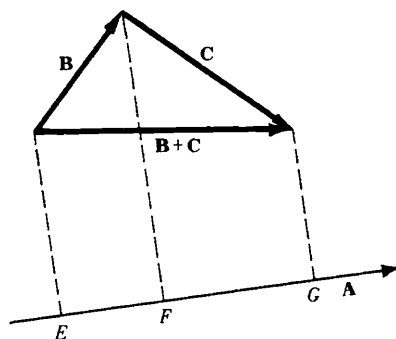
and

$$(\mathbf{B} + \mathbf{C}) \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{A} + \mathbf{C} \cdot \mathbf{A}$$

Then by the commutative law for dot products,

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$$

and the distributive law is valid.



4. Prove that
- $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{C} + \mathbf{D}) = \mathbf{A} \cdot \mathbf{C} + \mathbf{A} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{D}$
- .

By Problem 3, $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{C} + \mathbf{D}) = \mathbf{A} \cdot (\mathbf{C} + \mathbf{D}) + \mathbf{B} \cdot (\mathbf{C} + \mathbf{D}) = \mathbf{A} \cdot \mathbf{C} + \mathbf{A} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{D}$

The ordinary laws of algebra are valid for dot products.

5. Evaluate each of the following.

$$(a) \mathbf{i} \cdot \mathbf{i} = |\mathbf{i}| |\mathbf{i}| \cos 0^\circ = (1)(1)(1) = 1$$

$$(b) \mathbf{i} \cdot \mathbf{k} = |\mathbf{i}| |\mathbf{k}| \cos 90^\circ = (1)(1)(0) = 0$$

$$(c) \mathbf{k} \cdot \mathbf{j} = |\mathbf{k}| |\mathbf{j}| \cos 90^\circ = (1)(1)(0) = 0$$

$$(d) \mathbf{j} \cdot (2\mathbf{i} - 3\mathbf{j} + \mathbf{k}) = 2\mathbf{j} \cdot \mathbf{i} - 3\mathbf{j} \cdot \mathbf{j} + \mathbf{j} \cdot \mathbf{k} = 0 - 3 + 0 = -3$$

$$(e) (2\mathbf{i} - \mathbf{j}) \cdot (3\mathbf{i} + \mathbf{k}) = 2\mathbf{i} \cdot (3\mathbf{i} + \mathbf{k}) - \mathbf{j} \cdot (3\mathbf{i} + \mathbf{k}) = 6\mathbf{i} \cdot \mathbf{i} + 2\mathbf{i} \cdot \mathbf{k} - 3\mathbf{j} \cdot \mathbf{i} - \mathbf{j} \cdot \mathbf{k} = 6 + 0 - 0 - 0 = 6$$

6. If
- $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$
- and
- $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$
- , prove that
- $\mathbf{A} \cdot \mathbf{B} = A_1B_1 + A_2B_2 + A_3B_3$
- .

$$\mathbf{A} \cdot \mathbf{B} = (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \cdot (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k})$$

$$= A_1\mathbf{i} \cdot (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}) + A_2\mathbf{j} \cdot (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}) + A_3\mathbf{k} \cdot (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k})$$

$$= A_1B_1\mathbf{i} \cdot \mathbf{i} + A_1B_2\mathbf{i} \cdot \mathbf{j} + A_1B_3\mathbf{i} \cdot \mathbf{k} + A_2B_1\mathbf{j} \cdot \mathbf{i} + A_2B_2\mathbf{j} \cdot \mathbf{j} + A_2B_3\mathbf{j} \cdot \mathbf{k} + A_3B_1\mathbf{k} \cdot \mathbf{i} + A_3B_2\mathbf{k} \cdot \mathbf{j} + A_3B_3\mathbf{k} \cdot \mathbf{k}$$

$$= A_1B_1 + A_2B_2 + A_3B_3$$

since $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$ and all other dot products are zero.

7. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, show that $A = \sqrt{\mathbf{A} \cdot \mathbf{A}} = \sqrt{A_1^2 + A_2^2 + A_3^2}$.

$$\mathbf{A} \cdot \mathbf{A} = (A)(A) \cos 0^\circ = A^2. \quad \text{Then } A = \sqrt{\mathbf{A} \cdot \mathbf{A}}.$$

$$\begin{aligned} \text{Also, } \mathbf{A} \cdot \mathbf{A} &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \cdot (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \\ &= (A_1)(A_1) + (A_2)(A_2) + (A_3)(A_3) = A_1^2 + A_2^2 + A_3^2 \end{aligned}$$

by Problem 6, taking $\mathbf{B} = \mathbf{A}$.

$$\text{Then } A = \sqrt{\mathbf{A} \cdot \mathbf{A}} = \sqrt{A_1^2 + A_2^2 + A_3^2} \text{ is the magnitude of } \mathbf{A}. \text{ Sometimes } \mathbf{A} \cdot \mathbf{A} \text{ is written } A^2.$$

8. Find the angle between $\mathbf{A} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{B} = 6\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$.

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta, \quad A = \sqrt{(2)^2 + (2)^2 + (-1)^2} = 3, \quad B = \sqrt{(6)^2 + (-3)^2 + (2)^2} = 7$$

$$\mathbf{A} \cdot \mathbf{B} = (2)(6) + (2)(-3) + (-1)(2) = 12 - 6 - 2 = 4$$

$$\text{Then } \cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{4}{(3)(7)} = \frac{4}{21} = 0.1905 \quad \text{and} \quad \theta = 79^\circ \text{ approximately.}$$

9. If $\mathbf{A} \cdot \mathbf{B} = 0$ and if A and B are not zero, show that $\mathbf{A}$ is perpendicular to $\mathbf{B}$.

If $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta = 0$, then $\cos \theta = 0$ or $\theta = 90^\circ$. Conversely, if $\theta = 90^\circ$, $\mathbf{A} \cdot \mathbf{B} = 0$.

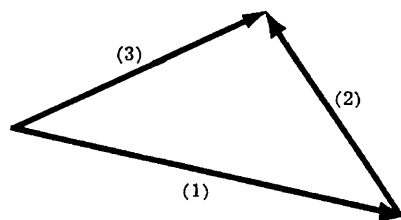
10. Determine the value of a so that $\mathbf{A} = 2\mathbf{i} + a\mathbf{j} + \mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ are perpendicular.

From Problem 9, $\mathbf{A}$ and $\mathbf{B}$ are perpendicular if $\mathbf{A} \cdot \mathbf{B} = 0$.

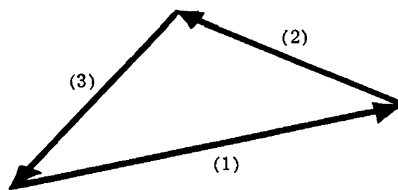
$$\text{Then } \mathbf{A} \cdot \mathbf{B} = (2)(4) + (a)(-2) + (1)(-2) = 8 - 2a - 2 = 0 \quad \text{for} \quad a = 3.$$

11. Show that the vectors $\mathbf{A} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{B} = \mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$, $\mathbf{C} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$ form a right triangle.

We first have to show that the vectors form a triangle.



(a)



(b)

From the figures it is seen that the vectors will form a triangle if

(a) one of the vectors, say (3), is the resultant or sum of (1) and (2),

(b) the sum or resultant of the vectors $(1) + (2) + (3)$ is zero,

according as (a) two vectors have a common terminal point or (b) none of the vectors have a common terminal point. By trial we find $\mathbf{A} = \mathbf{B} + \mathbf{C}$ so that the vectors do form a triangle.

Since $\mathbf{A} \cdot \mathbf{B} = (3)(1) + (-2)(-3) + (1)(5) = 14$, $\mathbf{A} \cdot \mathbf{C} = (3)(2) + (-2)(1) + (1)(-4) = 0$, and $\mathbf{B} \cdot \mathbf{C} = (1)(2) + (-3)(1) + (5)(-4) = -21$, it follows that $\mathbf{A}$ and $\mathbf{C}$ are perpendicular and the triangle is a right triangle.

12. Find the angles which the vector $\mathbf{A} = 3\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$ makes with the coordinate axes.

Let α, β, γ be the angles which $\mathbf{A}$ makes with the positive x, y, z axes respectively.

$$\mathbf{A} \cdot \mathbf{i} = (A)(1) \cos \alpha = \sqrt{(3)^2 + (-6)^2 + (2)^2} \cos \alpha = 7 \cos \alpha$$

$$\mathbf{A} \cdot \mathbf{i} = (3\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}) \cdot \mathbf{i} = 3\mathbf{i} \cdot \mathbf{i} - 6\mathbf{j} \cdot \mathbf{i} + 2\mathbf{k} \cdot \mathbf{i} = 3$$

Then $\cos \alpha = 3/7 = 0.4286$, and $\alpha = 64.6^\circ$ approximately.

Similarly, $\cos \beta = -6/7$, $\beta = 149^\circ$ and $\cos \gamma = 2/7$, $\gamma = 73.4^\circ$.

The cosines of α, β , and γ are called the *direction cosines* of $\mathbf{A}$. (See Prob. 27, Chap. 1).

13. Find the projection of the vector $\mathbf{A} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ on the vector $\mathbf{B} = 4\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$.

$$\text{A unit vector in the direction } \mathbf{B} \text{ is } \mathbf{b} = \frac{\mathbf{B}}{B} = \frac{4\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}}{\sqrt{(4)^2 + (-4)^2 + (7)^2}} = \frac{4}{9}\mathbf{i} - \frac{4}{9}\mathbf{j} + \frac{7}{9}\mathbf{k}.$$

$$\begin{aligned} \text{Projection of } \mathbf{A} \text{ on the vector } \mathbf{B} &= \mathbf{A} \cdot \mathbf{b} = (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot \left(\frac{4}{9}\mathbf{i} - \frac{4}{9}\mathbf{j} + \frac{7}{9}\mathbf{k}\right) \\ &= (1)\left(\frac{4}{9}\right) + (-2)\left(-\frac{4}{9}\right) + (1)\left(\frac{7}{9}\right) = \frac{19}{9}. \end{aligned}$$

14. Prove the law of cosines for plane triangles.

From Fig.(a) below, $\mathbf{B} + \mathbf{C} = \mathbf{A}$ or $\mathbf{C} = \mathbf{A} - \mathbf{B}$.

$$\text{Then } \mathbf{C} \cdot \mathbf{C} = (\mathbf{A} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = \mathbf{A} \cdot \mathbf{A} + \mathbf{B} \cdot \mathbf{B} - 2\mathbf{A} \cdot \mathbf{B}$$

$$\text{and } C^2 = A^2 + B^2 - 2AB \cos \theta.$$

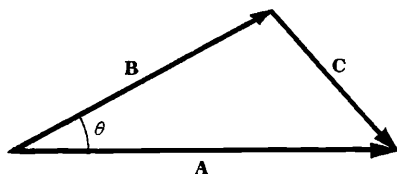


Fig.(a)

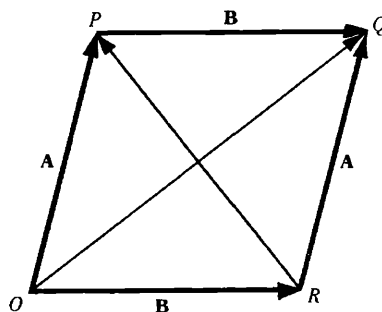


Fig.(b)

15. Prove that the diagonals of a rhombus are perpendicular. Refer to Fig.(b) above.

$$\mathbf{OQ} = \mathbf{OP} + \mathbf{PQ} = \mathbf{A} + \mathbf{B}$$

$$\mathbf{OR} + \mathbf{RP} = \mathbf{OP} \text{ or } \mathbf{B} + \mathbf{RP} = \mathbf{A} \text{ and } \mathbf{RP} = \mathbf{A} - \mathbf{B}$$

$$\text{Then } \mathbf{OQ} \cdot \mathbf{RP} = (\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = A^2 - B^2 = 0, \text{ since } A = B.$$

Hence $\mathbf{OQ}$ is perpendicular to $\mathbf{RP}$.

16. Determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

Let vector $\mathbf{C} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$ be perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. Then $\mathbf{C}$ is perpendicular to $\mathbf{A}$ and also to $\mathbf{B}$. Hence,

$$\mathbf{C} \cdot \mathbf{A} = 2c_1 - 6c_2 - 3c_3 = 0 \quad \text{or} \quad (1) \quad 2c_1 - 6c_2 = 3c_3$$

$$\mathbf{C} \cdot \mathbf{B} = 4c_1 + 3c_2 - c_3 = 0 \quad \text{or} \quad (2) \quad 4c_1 + 3c_2 = c_3$$

Solving (1) and (2) simultaneously: $c_1 = \frac{1}{2}c_3$, $c_2 = -\frac{1}{3}c_3$, $\mathbf{C} = c_3(\frac{1}{2}\mathbf{i} - \frac{1}{3}\mathbf{j} + \mathbf{k})$.

Then a unit vector in the direction of $\mathbf{C}$ is $\frac{\mathbf{C}}{C} = \frac{c_3(\frac{1}{2}\mathbf{i} - \frac{1}{3}\mathbf{j} + \mathbf{k})}{\sqrt{c_3^2[(\frac{1}{2})^2 + (-\frac{1}{3})^2 + (1)^2]}} = \pm(\frac{3}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k})$.

17. Find the work done in moving an object along a vector $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$ if the applied force is $\mathbf{F} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$. Refer to Fig.(a) below.

$$\begin{aligned}\text{Work done} &= (\text{magnitude of force in direction of motion})(\text{distance moved}) \\ &= (F \cos \theta)(r) = \mathbf{F} \cdot \mathbf{r} \\ &= (2\mathbf{i} - \mathbf{j} - \mathbf{k}) \cdot (3\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}) = 6 - 2 + 5 = 9.\end{aligned}$$

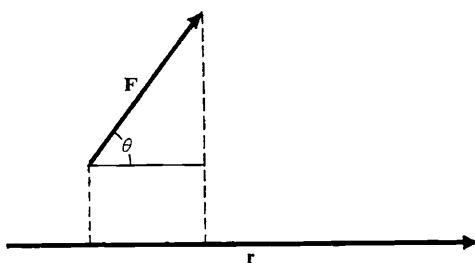


Fig.(a)

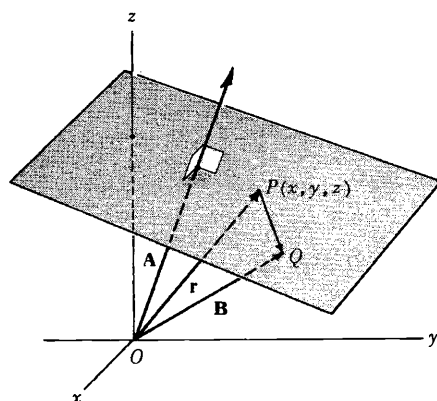


Fig.(b)

18. Find an equation for the plane perpendicular to the vector $\mathbf{A} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ and passing through the terminal point of the vector $\mathbf{B} = \mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$ (see Fig.(b) above).

Let $\mathbf{r}$ be the position vector of point P , and Q the terminal point of $\mathbf{B}$.

Since $\mathbf{PQ} = \mathbf{B} - \mathbf{r}$ is perpendicular to $\mathbf{A}$, $(\mathbf{B} - \mathbf{r}) \cdot \mathbf{A} = 0$ or $\mathbf{r} \cdot \mathbf{A} = \mathbf{B} \cdot \mathbf{A}$ is the required equation of the plane in vector form. In rectangular form this becomes

$$(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}) = (\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}) \cdot (2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})$$

$$\text{or} \quad 2x + 3y + 6z = (1)(2) + (5)(3) + (3)(6) = 35$$

19. In Problem 18 find the distance from the origin to the plane.

The distance from the origin to the plane is the projection of $\mathbf{B}$ on $\mathbf{A}$.

$$\text{A unit vector in direction } \mathbf{A} \text{ is } \mathbf{a} = \frac{\mathbf{A}}{A} = \frac{2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}}{\sqrt{(2)^2 + (3)^2 + (6)^2}} = \frac{2}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}.$$

$$\text{Then, projection of } \mathbf{B} \text{ on } \mathbf{A} = \mathbf{B} \cdot \mathbf{a} = (\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}) \cdot (\frac{2}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}) = 1(\frac{2}{7}) + 5(\frac{3}{7}) + 3(\frac{6}{7}) = 5.$$

20. If $\mathbf{A}$ is any vector, prove that $\mathbf{A} = (\mathbf{A} \cdot \mathbf{i})\mathbf{i} + (\mathbf{A} \cdot \mathbf{j})\mathbf{j} + (\mathbf{A} \cdot \mathbf{k})\mathbf{k}$.

$$\text{Since } \mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}, \quad \mathbf{A} \cdot \mathbf{i} = A_1\mathbf{i} \cdot \mathbf{i} + A_2\mathbf{j} \cdot \mathbf{i} + A_3\mathbf{k} \cdot \mathbf{i} = A_1$$

$$\text{Similarly, } \mathbf{A} \cdot \mathbf{j} = A_2 \text{ and } \mathbf{A} \cdot \mathbf{k} = A_3.$$

$$\text{Then } \mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k} = (\mathbf{A} \cdot \mathbf{i})\mathbf{i} + (\mathbf{A} \cdot \mathbf{j})\mathbf{j} + (\mathbf{A} \cdot \mathbf{k})\mathbf{k}.$$

THE CROSS OR VECTOR PRODUCT.

21. Prove $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$.

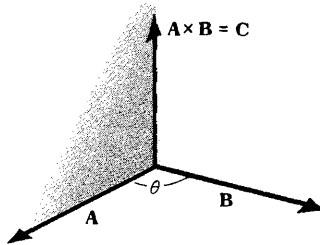


Fig.(a)

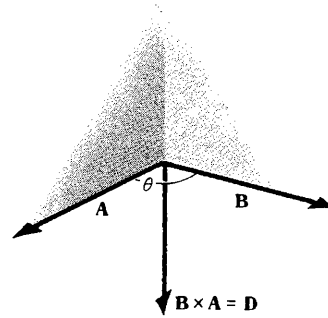


Fig.(b)

$\mathbf{A} \times \mathbf{B} = \mathbf{C}$ has magnitude $AB \sin \theta$ and direction such that $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ form a right-handed system (Fig.(a) above).

$\mathbf{B} \times \mathbf{A} = \mathbf{D}$ has magnitude $BA \sin \theta$ and direction such that $\mathbf{B}, \mathbf{A}$ and $\mathbf{D}$ form a right-handed system (Fig.(b) above).

Then $\mathbf{D}$ has the same magnitude as $\mathbf{C}$ but is opposite in direction, i.e. $\mathbf{C} = -\mathbf{D}$ or $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$.

The commutative law for cross products is not valid.

22. If $\mathbf{A} \times \mathbf{B} = \mathbf{0}$ and if $\mathbf{A}$ and $\mathbf{B}$ are not zero, show that $\mathbf{A}$ is parallel to $\mathbf{B}$.

If $\mathbf{A} \times \mathbf{B} = AB \sin \theta \mathbf{u} = \mathbf{0}$, then $\sin \theta = 0$ and $\theta = 0^\circ$ or 180° .

23. Show that $|\mathbf{A} \times \mathbf{B}|^2 + |\mathbf{A} \cdot \mathbf{B}|^2 = |\mathbf{A}|^2 |\mathbf{B}|^2$.

$$|\mathbf{A} \times \mathbf{B}|^2 + |\mathbf{A} \cdot \mathbf{B}|^2 = |AB \sin \theta \mathbf{u}|^2 + |AB \cos \theta|^2 = A^2 B^2 \sin^2 \theta + A^2 B^2 \cos^2 \theta \\ = A^2 B^2 = |\mathbf{A}|^2 |\mathbf{B}|^2$$

24. Evaluate each of the following.

(a) $\mathbf{i} \times \mathbf{j} = \mathbf{k}$

(f) $\mathbf{j} \times \mathbf{j} = \mathbf{0}$

(b) $\mathbf{j} \times \mathbf{k} = \mathbf{i}$

(g) $\mathbf{i} \times \mathbf{k} = -\mathbf{k} \times \mathbf{i} = -\mathbf{j}$

(c) $\mathbf{k} \times \mathbf{i} = \mathbf{j}$

(h) $(2\mathbf{j}) \times (3\mathbf{k}) = 6 \mathbf{j} \times \mathbf{k} = 6\mathbf{i}$

(d) $\mathbf{k} \times \mathbf{j} = -\mathbf{j} \times \mathbf{k} = -\mathbf{i}$

(i) $(3\mathbf{l}) \times (-2\mathbf{k}) = -6 \mathbf{i} \times \mathbf{k} = 6\mathbf{j}$

(e) $\mathbf{i} \times \mathbf{i} = \mathbf{0}$

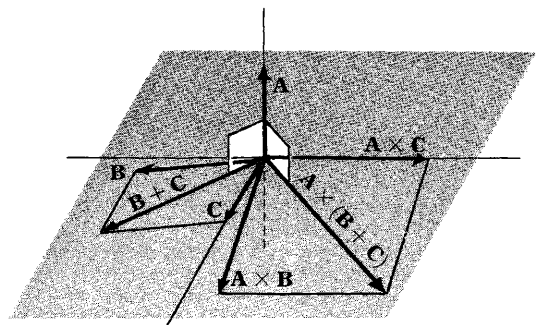
(j) $2\mathbf{j} \times \mathbf{i} - 3\mathbf{k} = -2\mathbf{k} - 3\mathbf{k} = -5\mathbf{k}$

25. Prove that $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$ for the case where $\mathbf{A}$ is perpendicular to $\mathbf{B}$ and also to $\mathbf{C}$.

Since $\mathbf{A}$ is perpendicular to $\mathbf{B}$, $\mathbf{A} \times \mathbf{B}$ is a vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$ and having magnitude $AB \sin 90^\circ = AB$ or magnitude of $\mathbf{AB}$. This is equivalent to multiplying vector $\mathbf{B}$ by A and rotating the resultant vector through 90° to the position shown in the adjoining diagram.

Similarly, $\mathbf{A} \times \mathbf{C}$ is the vector obtained by multiplying $\mathbf{C}$ by A and rotating the resultant vector through 90° to the position shown.

In like manner, $\mathbf{A} \times (\mathbf{B} + \mathbf{C})$ is the vector obtained



by multiplying $\mathbf{B} + \mathbf{C}$ by A and rotating the resultant vector through 90° to the position shown.

Since $\mathbf{A} \times (\mathbf{B} + \mathbf{C})$ is the diagonal of the parallelogram with $\mathbf{A} \times \mathbf{B}$ and $\mathbf{A} \times \mathbf{C}$ as sides, we have $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$.

26. Prove that $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$ in the general case where $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are non-coplanar.

Resolve $\mathbf{B}$ into two component vectors, one perpendicular to $\mathbf{A}$ and the other parallel to $\mathbf{A}$, and denote them by $\mathbf{B}_\perp$ and $\mathbf{B}_\parallel$, respectively. Then $\mathbf{B} = \mathbf{B}_\perp + \mathbf{B}_\parallel$.

If θ is the angle between $\mathbf{A}$ and $\mathbf{B}$, then $B_\perp = B \sin \theta$. Thus the magnitude of $\mathbf{A} \times \mathbf{B}_\perp$ is $AB \sin \theta$, the same as the magnitude of $\mathbf{A} \times \mathbf{B}$. Also, the direction of $\mathbf{A} \times \mathbf{B}_\perp$ is the same as the direction of $\mathbf{A} \times \mathbf{B}$. Hence $\mathbf{A} \times \mathbf{B}_\perp = \mathbf{A} \times \mathbf{B}$.

Similarly if $\mathbf{C}$ is resolved into two component vectors $\mathbf{C}_\parallel$ and $\mathbf{C}_\perp$, parallel and perpendicular respectively to $\mathbf{A}$, then $\mathbf{A} \times \mathbf{C}_\perp = \mathbf{A} \times \mathbf{C}$.

Also, since $\mathbf{B} + \mathbf{C} = \mathbf{B}_\perp + \mathbf{B}_\parallel + \mathbf{C}_\perp + \mathbf{C}_\parallel = (\mathbf{B}_\perp + \mathbf{C}_\perp) + (\mathbf{B}_\parallel + \mathbf{C}_\parallel)$ it follows that

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times (\mathbf{B} + \mathbf{C}).$$

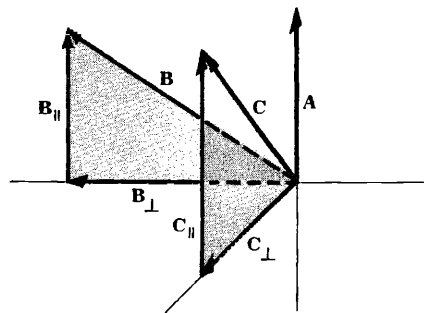
Now $\mathbf{B}_\perp$ and $\mathbf{C}_\perp$ are vectors perpendicular to $\mathbf{A}$ and so by Problem 25,

$$\mathbf{A} \times (\mathbf{B}_\perp + \mathbf{C}_\perp) = \mathbf{A} \times \mathbf{B}_\perp + \mathbf{A} \times \mathbf{C}_\perp$$

Then

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$$

and the distributive law holds. Multiplying by -1 , using Prob. 21, this becomes $(\mathbf{B} + \mathbf{C}) \times \mathbf{A} = \mathbf{B} \times \mathbf{A} + \mathbf{C} \times \mathbf{A}$. Note that the order of factors in cross products is important. The usual laws of algebra apply only if proper order is maintained.



27. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, prove that $\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$.

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \times (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}) \\ &= A_1\mathbf{i} \times (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}) + A_2\mathbf{j} \times (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}) + A_3\mathbf{k} \times (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}) \\ &= A_1B_1\mathbf{i} \times \mathbf{i} + A_1B_2\mathbf{i} \times \mathbf{j} + A_1B_3\mathbf{i} \times \mathbf{k} + A_2B_1\mathbf{j} \times \mathbf{i} + A_2B_2\mathbf{j} \times \mathbf{j} + A_2B_3\mathbf{j} \times \mathbf{k} + A_3B_1\mathbf{k} \times \mathbf{i} + A_3B_2\mathbf{k} \times \mathbf{j} + A_3B_3\mathbf{k} \times \mathbf{k} \\ &= (A_2B_3 - A_3B_2)\mathbf{i} + (A_3B_1 - A_1B_3)\mathbf{j} + (A_1B_2 - A_2B_1)\mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}. \end{aligned}$$

28. If $\mathbf{A} = 2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ and $\mathbf{B} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, find (a) $\mathbf{A} \times \mathbf{B}$, (b) $\mathbf{B} \times \mathbf{A}$, (c) $(\mathbf{A} + \mathbf{B}) \times (\mathbf{A} - \mathbf{B})$.

$$\begin{aligned} \text{(a) } \mathbf{A} \times \mathbf{B} &= (2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \times (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -3 & -1 \\ 1 & 4 & -2 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} -3 & -1 \\ 4 & -2 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 2 & -3 \\ 1 & 4 \end{vmatrix} = 10\mathbf{i} + 3\mathbf{j} + 11\mathbf{k} \end{aligned}$$

Another Method.

$$\begin{aligned} (2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \times (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) &= 2\mathbf{i} \times (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) - 3\mathbf{j} \times (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) - \mathbf{k} \times (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) \\ &= 2\mathbf{i} \times \mathbf{i} + 8\mathbf{i} \times \mathbf{j} - 4\mathbf{i} \times \mathbf{k} - 3\mathbf{j} \times \mathbf{i} - 12\mathbf{j} \times \mathbf{j} + 6\mathbf{j} \times \mathbf{k} - \mathbf{k} \times \mathbf{i} - 4\mathbf{k} \times \mathbf{j} + 2\mathbf{k} \times \mathbf{k} \\ &= \mathbf{0} + 8\mathbf{k} + 4\mathbf{j} + 3\mathbf{k} - \mathbf{0} + 6\mathbf{i} - \mathbf{j} + 4\mathbf{i} + \mathbf{0} = 10\mathbf{i} + 3\mathbf{j} + 11\mathbf{k} \end{aligned}$$

$$\begin{aligned}
 (b) \quad \mathbf{B} \times \mathbf{A} &= (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) \times (2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 4 & -2 \\ 2 & -3 & -1 \end{vmatrix} \\
 &= \mathbf{i} \begin{vmatrix} 4 & -2 \\ -3 & -1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & -2 \\ 2 & -1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & 4 \\ 2 & -3 \end{vmatrix} = -10\mathbf{i} - 3\mathbf{j} - 11\mathbf{k}.
 \end{aligned}$$

Comparing with (a), $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$. Note that this is equivalent to the theorem: If two rows of a determinant are interchanged, the determinant changes sign.

$$(c) \quad \mathbf{A} + \mathbf{B} = (2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) + (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) = 3\mathbf{i} + \mathbf{j} - 3\mathbf{k}$$

$$\mathbf{A} - \mathbf{B} = (2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) - (\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) = \mathbf{i} - 7\mathbf{j} + \mathbf{k}$$

$$\begin{aligned}
 \text{Then } (\mathbf{A} + \mathbf{B}) \times (\mathbf{A} - \mathbf{B}) &= (3\mathbf{i} + \mathbf{j} - 3\mathbf{k}) \times (\mathbf{i} - 7\mathbf{j} + \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -3 \\ 1 & -7 & 1 \end{vmatrix} \\
 &= \mathbf{i} \begin{vmatrix} 1 & -3 \\ -7 & 1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 3 & -3 \\ 1 & 1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 3 & 1 \\ 1 & -7 \end{vmatrix} = -20\mathbf{i} - 6\mathbf{j} - 22\mathbf{k}.
 \end{aligned}$$

Another Method.

$$\begin{aligned}
 (\mathbf{A} + \mathbf{B}) \times (\mathbf{A} - \mathbf{B}) &= \mathbf{A} \times (\mathbf{A} - \mathbf{B}) + \mathbf{B} \times (\mathbf{A} - \mathbf{B}) \\
 &= \mathbf{A} \times \mathbf{A} - \mathbf{A} \times \mathbf{B} + \mathbf{B} \times \mathbf{A} - \mathbf{B} \times \mathbf{B} = \mathbf{0} - \mathbf{A} \times \mathbf{B} - \mathbf{A} \times \mathbf{B} - \mathbf{0} = -2\mathbf{A} \times \mathbf{B} \\
 &= -2(10\mathbf{i} + 3\mathbf{j} + 11\mathbf{k}) = -20\mathbf{i} - 6\mathbf{j} - 22\mathbf{k}, \text{ using (a).}
 \end{aligned}$$

29. If $\mathbf{A} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$, and $\mathbf{C} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, find (a) $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$, (b) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

$$(a) \quad \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix} = -\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}.$$

$$\text{Then } (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (-\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}) \times (\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 7 & 5 \\ 1 & -2 & 2 \end{vmatrix} = 24\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}.$$

$$(b) \quad \mathbf{B} \times \mathbf{C} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -1 \\ 1 & -2 & 2 \end{vmatrix} = 0\mathbf{i} - 5\mathbf{j} - 5\mathbf{k} = -5\mathbf{j} - 5\mathbf{k}.$$

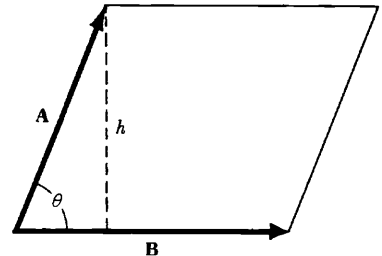
$$\text{Then } \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (3\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \times (-5\mathbf{j} - 5\mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 2 \\ 0 & -5 & -5 \end{vmatrix} = 15\mathbf{i} + 15\mathbf{j} - 15\mathbf{k}.$$

Thus $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \neq \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$, showing the need for parentheses in $\mathbf{A} \times \mathbf{B} \times \mathbf{C}$ to avoid ambiguity.

30. Prove that the area of a parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$ is $|\mathbf{A} \times \mathbf{B}|$.

$$\begin{aligned}
 \text{Area of parallelogram} &= h |\mathbf{B}| \\
 &= |\mathbf{A}| \sin \theta |\mathbf{B}| \\
 &= |\mathbf{A} \times \mathbf{B}|.
 \end{aligned}$$

Note that the area of the triangle with sides $\mathbf{A}$ and $\mathbf{B}$ is $\frac{1}{2} |\mathbf{A} \times \mathbf{B}|$.



31. Find the area of the triangle having vertices at $P(1, 3, 2)$, $Q(2, -1, 1)$, $R(-1, 2, 3)$.

$$\mathbf{PQ} = (2-1)\mathbf{i} + (-1-3)\mathbf{j} + (1-2)\mathbf{k} = \mathbf{i} - 4\mathbf{j} - \mathbf{k}$$

$$\mathbf{PR} = (-1-1)\mathbf{i} + (2-3)\mathbf{j} + (3-2)\mathbf{k} = -2\mathbf{i} - \mathbf{j} + \mathbf{k}$$

From Problem 30,

$$\begin{aligned} \text{area of triangle} &= \frac{1}{2} |\mathbf{PQ} \times \mathbf{PR}| = \frac{1}{2} |(\mathbf{i} - 4\mathbf{j} - \mathbf{k}) \times (-2\mathbf{i} - \mathbf{j} + \mathbf{k})| \\ &= \frac{1}{2} \left| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -4 & -1 \\ -2 & -1 & 1 \end{vmatrix} \right| = \frac{1}{2} |-5\mathbf{i} + \mathbf{j} - 9\mathbf{k}| = \frac{1}{2} \sqrt{(-5)^2 + (1)^2 + (-9)^2} = \frac{1}{2} \sqrt{107}. \end{aligned}$$

32. Determine a unit vector perpendicular to the plane of $\mathbf{A} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

$\mathbf{A} \times \mathbf{B}$ is a vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$.

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -6 & -3 \\ 4 & 3 & -1 \end{vmatrix} = 15\mathbf{i} - 10\mathbf{j} + 30\mathbf{k}$$

A unit vector parallel to $\mathbf{A} \times \mathbf{B}$ is $\frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = \frac{15\mathbf{i} - 10\mathbf{j} + 30\mathbf{k}}{\sqrt{(15)^2 + (-10)^2 + (30)^2}} = \frac{3}{7}\mathbf{i} - \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}.$

Another unit vector, opposite in direction, is $(-3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k})/7$.

Compare with Problem 16.

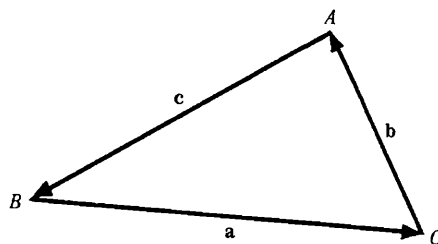
33. Prove the law of sines for plane triangles.

Let $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ represent the sides of triangle ABC as shown in the adjoining figure; then $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$. Multiplying by $\mathbf{a} \times$, $\mathbf{b} \times$ and $\mathbf{c} \times$ in succession, we find

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$$

i.e. $ab \sin C = bc \sin A = ca \sin B$

or $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}.$



34. Consider a tetrahedron with faces F_1, F_2, F_3, F_4 . Let $\mathbf{V}_1, \mathbf{V}_2, \mathbf{V}_3, \mathbf{V}_4$ be vectors whose magnitudes are respectively equal to the areas of F_1, F_2, F_3, F_4 and whose directions are perpendicular to these faces in the outward direction. Show that $\mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 + \mathbf{V}_4 = \mathbf{0}$.

By Problem 30, the area of a triangular face determined by $\mathbf{R}$ and $\mathbf{S}$ is $\frac{1}{2} |\mathbf{R} \times \mathbf{S}|$.

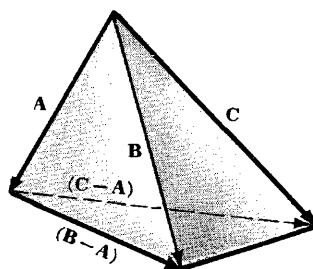
The vectors associated with each of the faces of the tetrahedron are

$$\mathbf{V}_1 = \frac{1}{2} \mathbf{A} \times \mathbf{B}, \quad \mathbf{V}_2 = \frac{1}{2} \mathbf{B} \times \mathbf{C}, \quad \mathbf{V}_3 = \frac{1}{2} \mathbf{C} \times \mathbf{A}, \quad \mathbf{V}_4 = \frac{1}{2} (\mathbf{C} - \mathbf{A}) \times (\mathbf{B} - \mathbf{A})$$

$$\begin{aligned} \text{Then } \mathbf{V}_1 + \mathbf{V}_2 + \mathbf{V}_3 + \mathbf{V}_4 &= \frac{1}{2} [\mathbf{A} \times \mathbf{B} + \mathbf{B} \times \mathbf{C} + \mathbf{C} \times \mathbf{A} + (\mathbf{C} - \mathbf{A}) \times (\mathbf{B} - \mathbf{A})] \\ &= \frac{1}{2} [\mathbf{A} \times \mathbf{B} + \mathbf{B} \times \mathbf{C} + \mathbf{C} \times \mathbf{A} + \mathbf{C} \times \mathbf{B} - \mathbf{C} \times \mathbf{A} - \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{A}] = \mathbf{0}. \end{aligned}$$

This result can be generalized to closed polyhedra and in the limiting case to any closed surface.

Because of the application presented here it is sometimes convenient to assign a direction to area and we speak of the *vector area*.



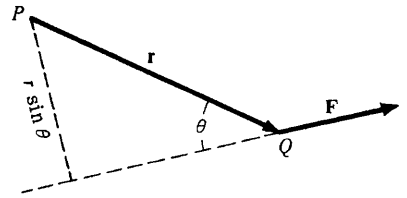
35. Find an expression for the moment of a force $\mathbf{F}$ about a point P .

The moment $\mathbf{M}$ of $\mathbf{F}$ about P is in magnitude equal to F times the perpendicular distance from P to the

line of action of $\mathbf{F}$. Then if $\mathbf{r}$ is the vector from P to the initial point Q of $\mathbf{F}$,

$$M = F(r \sin \theta) = rF \sin \theta = |\mathbf{r} \times \mathbf{F}|$$

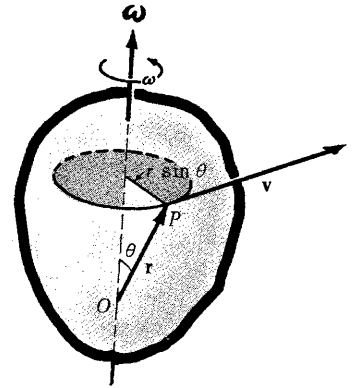
If we think of a right-threaded screw at P perpendicular to the plane of $\mathbf{r}$ and $\mathbf{F}$, then when the force $\mathbf{F}$ acts the screw will move in the direction of $\mathbf{r} \times \mathbf{F}$. Because of this it is convenient to define the moment as the vector $\mathbf{M} = \mathbf{r} \times \mathbf{F}$.



36. A rigid body rotates about an axis through point O with angular speed ω . Prove that the linear velocity $\mathbf{v}$ of a point P of the body with position vector $\mathbf{r}$ is given by $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, where $\boldsymbol{\omega}$ is the vector with magnitude ω whose direction is that in which a right-handed screw would advance under the given rotation.

Since P travels in a circle of radius $r \sin \theta$, the magnitude of the linear velocity $\mathbf{v}$ is $\omega(r \sin \theta) = |\boldsymbol{\omega} \times \mathbf{r}|$. Also, $\mathbf{v}$ must be perpendicular to both $\boldsymbol{\omega}$ and $\mathbf{r}$ and is such that $\mathbf{r}, \boldsymbol{\omega}$ and $\mathbf{v}$ form a right-handed system.

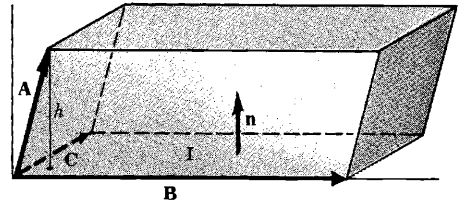
Then $\mathbf{v}$ agrees both in magnitude and direction with $\boldsymbol{\omega} \times \mathbf{r}$; hence $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$. The vector $\boldsymbol{\omega}$ is called the *angular velocity*.



TRIPLE PRODUCTS.

37. Show that $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ is in absolute value equal to the volume of a parallelepiped with sides $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$.

Let $\mathbf{n}$ be a unit normal to parallelogram I , having the direction of $\mathbf{B} \times \mathbf{C}$, and let h be the height of the terminal point of $\mathbf{A}$ above the parallelogram I .



$$\begin{aligned} \text{Volume of parallelepiped} &= (\text{height } h)(\text{area of parallelogram } I) \\ &= (\mathbf{A} \cdot \mathbf{n})(|\mathbf{B} \times \mathbf{C}|) \\ &= \mathbf{A} \cdot \{|\mathbf{B} \times \mathbf{C}| \mathbf{n}\} = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \end{aligned}$$

If $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ do not form a right-handed system, $\mathbf{A} \cdot \mathbf{n} < 0$ and the volume = $|\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})|$.

38. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$ show that

$$\begin{aligned} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \\ \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{A} \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \\ &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \cdot [(B_2C_3 - B_3C_2)\mathbf{i} + (B_3C_1 - B_1C_3)\mathbf{j} + (B_1C_2 - B_2C_1)\mathbf{k}] \\ &= A_1(B_2C_3 - B_3C_2) + A_2(B_3C_1 - B_1C_3) + A_3(B_1C_2 - B_2C_1) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \end{aligned}$$

39. Evaluate $(2\mathbf{i} - 3\mathbf{j}) \cdot [(\mathbf{i} + \mathbf{j} - \mathbf{k}) \times (3\mathbf{i} - \mathbf{k})]$.

By Problem 38, the result is $\begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix} = 4.$

Another Method. The result is equal to

$$\begin{aligned} & (2\mathbf{i} - 3\mathbf{j}) \cdot [\mathbf{i} \times (3\mathbf{i} - \mathbf{k}) + \mathbf{j} \times (3\mathbf{i} - \mathbf{k}) - \mathbf{k} \times (3\mathbf{i} - \mathbf{k})] \\ &= (2\mathbf{i} - 3\mathbf{j}) \cdot [3\mathbf{i} \times \mathbf{i} - \mathbf{i} \times \mathbf{k} + 3\mathbf{j} \times \mathbf{i} - \mathbf{j} \times \mathbf{k} - 3\mathbf{k} \times \mathbf{i} + \mathbf{k} \times \mathbf{k}] \\ &= (2\mathbf{i} - 3\mathbf{j}) \cdot (0 + \mathbf{j} - 3\mathbf{k} - \mathbf{i} - 3\mathbf{j} + 0) \\ &= (2\mathbf{i} - 3\mathbf{j}) \cdot (-\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}) = (2)(-1) + (-3)(-2) + (0)(-3) = 4. \end{aligned}$$

40. Prove that $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}).$

By Problem 38, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}.$

By a theorem of determinants which states that interchange of two rows of a determinant changes its sign, we have

$$\begin{aligned} \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} &= - \begin{vmatrix} B_1 & B_2 & B_3 \\ A_1 & A_2 & A_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = \begin{vmatrix} B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) \\ \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} &= - \begin{vmatrix} C_1 & C_2 & C_3 \\ B_1 & B_2 & B_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = \begin{vmatrix} C_1 & C_2 & C_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \end{aligned}$$

41. Show that $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}.$

From Problem 40, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$

Occasionally $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ is written without parentheses as $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$. In such case there cannot be any ambiguity since the only possible interpretations are $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ and $(\mathbf{A} \cdot \mathbf{B}) \times \mathbf{C}$. The latter however has no meaning since the cross product of a scalar with a vector is undefined.

The result $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$ is sometimes summarized in the statement that the dot and cross can be interchanged without affecting the result.

42. Prove that $\mathbf{A} \cdot (\mathbf{A} \times \mathbf{C}) = 0.$

From Problem 41, $\mathbf{A} \cdot (\mathbf{A} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{A}) \cdot \mathbf{C} = 0.$

43. Prove that a necessary and sufficient condition for the vectors $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ to be coplanar is that $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = 0.$

Note that $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$ can have no meaning other than $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$.

If $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ are coplanar the volume of the parallelepiped formed by them is zero. Then by Problem 37, $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = 0.$

Conversely, if $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = 0$ the volume of the parallelepiped formed by vectors $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ is zero, and so the vectors must lie in a plane.

44. Let $\mathbf{r}_1 = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$, $\mathbf{r}_2 = x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}$ and $\mathbf{r}_3 = x_3\mathbf{i} + y_3\mathbf{j} + z_3\mathbf{k}$ be the position vectors of

points $P_1(x_1, y_1, z_1)$, $P_2(x_2, y_2, z_2)$ and $P_3(x_3, y_3, z_3)$. Find an equation for the plane passing through P_1 , P_2 and P_3 .

We assume that P_1, P_2 and P_3 do not lie in the same straight line; hence they determine a plane.

Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ denote the position vector of any point $P(x, y, z)$ in the plane. Consider vectors $\mathbf{P}_1\mathbf{P}_2 = \mathbf{r}_2 - \mathbf{r}_1$, $\mathbf{P}_1\mathbf{P}_3 = \mathbf{r}_3 - \mathbf{r}_1$ and $\mathbf{P}_1\mathbf{P} = \mathbf{r} - \mathbf{r}_1$ which all lie in the plane.

By Problem 43, $\mathbf{P}_1\mathbf{P} \cdot \mathbf{P}_1\mathbf{P}_2 \times \mathbf{P}_1\mathbf{P}_3 = 0$ or

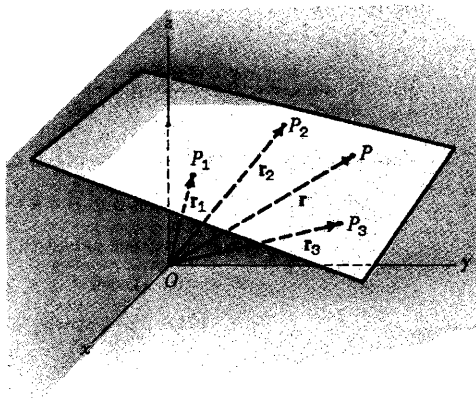
$$(\mathbf{r} - \mathbf{r}_1) \cdot (\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1) = 0$$

In terms of rectangular coordinates this becomes

$$[(x-x_1)\mathbf{i} + (y-y_1)\mathbf{j} + (z-z_1)\mathbf{k}] \cdot [(x_2-x_1)\mathbf{i} + (y_2-y_1)\mathbf{j} + (z_2-z_1)\mathbf{k}] \times [(x_3-x_1)\mathbf{i} + (y_3-y_1)\mathbf{j} + (z_3-z_1)\mathbf{k}] = 0$$

or, using Problem 38,

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0.$$



45. Find an equation for the plane determined by the points $P_1(2, -1, 1)$, $P_2(3, 2, -1)$ and $P_3(-1, 3, 2)$.

The position vectors of P_1, P_2, P_3 and any point $P(x, y, z)$ are respectively $\mathbf{r}_1 = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{r}_2 = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{r}_3 = -\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

Then $\mathbf{P}\mathbf{P}_1 = \mathbf{r} - \mathbf{r}_1$, $\mathbf{P}_2\mathbf{P}_1 = \mathbf{r}_2 - \mathbf{r}_1$, $\mathbf{P}_3\mathbf{P}_1 = \mathbf{r}_3 - \mathbf{r}_1$ all lie in the required plane, so that

$$(\mathbf{r} - \mathbf{r}_1) \cdot (\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1) = 0$$

i.e. $[(x-2)\mathbf{i} + (y+1)\mathbf{j} + (z-1)\mathbf{k}] \cdot [\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}] \times [-3\mathbf{i} + 4\mathbf{j} + \mathbf{k}] = 0$

$$[(x-2)\mathbf{i} + (y+1)\mathbf{j} + (z-1)\mathbf{k}] \cdot [11\mathbf{i} + 5\mathbf{j} + 13\mathbf{k}] = 0$$

$$11(x-2) + 5(y+1) + 13(z-1) = 0 \quad \text{or} \quad 11x + 5y + 13z = 30.$$

46. If the points P, Q and R , not all lying on the same straight line, have position vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ relative to a given origin, show that $\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}$ is a vector perpendicular to the plane of P, Q and R .

Let $\mathbf{r}$ be the position vector of any point in the plane of P, Q and R . Then the vectors $\mathbf{r} - \mathbf{a}$, $\mathbf{b} - \mathbf{a}$ and $\mathbf{c} - \mathbf{a}$ are coplanar, so that by Problem 43

$$(\mathbf{r} - \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) = 0 \quad \text{or} \quad (\mathbf{r} - \mathbf{a}) \cdot (\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}) = 0.$$

Thus $\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}$ is perpendicular to $\mathbf{r} - \mathbf{a}$ and is therefore perpendicular to the plane of P, Q and R .

47. Prove: (a) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$, (b) $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$.

(a) Let $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$.

$$\begin{aligned} \text{Then } \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \times \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} \\ &= (A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}) \times ([B_2C_3 - B_3C_2]\mathbf{i} + [B_3C_1 - B_1C_3]\mathbf{j} + [B_1C_2 - B_2C_1]\mathbf{k}) \end{aligned}$$

$$\begin{aligned}
&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_2C_3 - B_3C_2 & B_3C_1 - B_1C_3 & B_1C_2 - B_2C_1 \end{vmatrix} \\
&= (A_2B_1C_2 - A_2B_2C_1 - A_3B_3C_1 + A_3B_1C_3)\mathbf{i} + (A_3B_2C_3 - A_3B_3C_2 - A_1B_1C_2 + A_1B_2C_1)\mathbf{j} \\
&\quad + (A_1B_3C_1 - A_1B_1C_3 - A_2B_2C_3 + A_2B_3C_2)\mathbf{k}
\end{aligned}$$

Also $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

$$\begin{aligned}
&= (B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k})(A_1C_1 + A_2C_2 + A_3C_3) - (C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k})(A_1B_1 + A_2B_2 + A_3B_3) \\
&= (A_2B_1C_2 + A_3B_1C_3 - A_2C_1B_2 - A_3C_1B_3)\mathbf{i} + (B_2A_1C_1 + B_2A_3C_3 - C_2A_1B_1 - C_2A_3B_3)\mathbf{j} \\
&\quad + (B_3A_1C_1 + B_3A_2C_2 - C_3A_1B_1 - C_3A_2B_2)\mathbf{k}
\end{aligned}$$

and the result follows.

(b) $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = -\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = -\{\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A})\} = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$ upon replacing $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ in (a) by $\mathbf{C}, \mathbf{A}$ and $\mathbf{B}$ respectively.

Note that $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$, i.e. the associative law for vector cross products is not valid for all vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$.

48. Prove: $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C})$.

From Problem 41, $\mathbf{X} \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{X} \times \mathbf{C}) \cdot \mathbf{D}$. Let $\mathbf{X} = \mathbf{A} \times \mathbf{B}$; then

$$\begin{aligned}
(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) &= \{(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}\} \cdot \mathbf{D} = \{\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})\} \cdot \mathbf{D} \\
&= (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}), \quad \text{using Problem 47(b).}
\end{aligned}$$

49. Prove: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{0}$.

By Problem 47(a),

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$\mathbf{B} \times (\mathbf{C} \times \mathbf{A}) = \mathbf{C}(\mathbf{B} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$$

$$\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$$

Adding, the result follows.

50. Prove: $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C} \times \mathbf{D}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{D}) - \mathbf{D}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})$.

By Problem 47(a), $\mathbf{X} \times (\mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{X} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{X} \cdot \mathbf{C})$. Let $\mathbf{X} = \mathbf{A} \times \mathbf{B}$; then

$$\begin{aligned}
(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) &= \mathbf{C}(\mathbf{A} \times \mathbf{B} \cdot \mathbf{D}) - \mathbf{D}(\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}) \\
&= \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{D}) - \mathbf{D}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})
\end{aligned}$$

By Problem 47(b), $(\mathbf{A} \times \mathbf{B}) \times \mathbf{Y} = \mathbf{B}(\mathbf{A} \cdot \mathbf{Y}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{Y})$. Let $\mathbf{Y} = \mathbf{C} \times \mathbf{D}$; then

$$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C} \times \mathbf{D}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{D})$$

51. Let PQR be a spherical triangle whose sides p, q, r are arcs of great circles. Prove that

$$\frac{\sin P}{\sin p} = \frac{\sin Q}{\sin q} = \frac{\sin R}{\sin r}$$

Suppose that the sphere (see figure below) has unit radius, and let unit vectors $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ be drawn from the center O of the sphere to P, Q and R respectively. From Problem 50,

$$(1) \quad (\mathbf{A} \times \mathbf{B}) \times (\mathbf{A} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) \mathbf{A}$$

A unit vector perpendicular to $\mathbf{A} \times \mathbf{B}$ and $\mathbf{A} \times \mathbf{C}$ is $\mathbf{A}$, so that (1) becomes

$$(2) \quad \sin r \sin q \sin P \mathbf{A} = (\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) \mathbf{A} \quad \text{or}$$

$$(3) \quad \sin r \sin q \sin P = \mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$$

By cyclic permutation of p, q, r, P, Q, R and $\mathbf{A}, \mathbf{B}, \mathbf{C}$ we obtain

$$(4) \quad \sin p \sin r \sin Q = \mathbf{B} \cdot \mathbf{C} \times \mathbf{A}$$

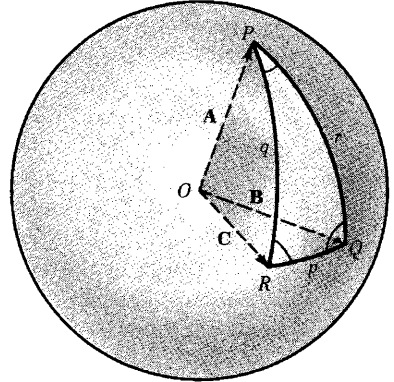
$$(5) \quad \sin q \sin p \sin R = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B}$$

Then since the right hand sides of (3), (4) and (5) are equal (Problem 40)

$$\sin r \sin q \sin P = \sin p \sin r \sin Q = \sin q \sin p \sin R$$

$$\text{from which we find} \quad \frac{\sin P}{\sin p} = \frac{\sin Q}{\sin q} = \frac{\sin R}{\sin r}$$

This is called the *law of sines* for spherical triangles.



$$52. \text{ Prove: } (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) = (\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})^2.$$

By Problem 47(a), $\mathbf{X} \times (\mathbf{C} \times \mathbf{A}) = \mathbf{C}(\mathbf{X} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{X} \cdot \mathbf{C})$. Let $\mathbf{X} = \mathbf{B} \times \mathbf{C}$; then

$$\begin{aligned} (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) &= \mathbf{C}(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{B} \times \mathbf{C} \cdot \mathbf{C}) \\ &= \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{C}) = \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) \end{aligned}$$

$$\begin{aligned} \text{Thus } (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) &= (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) \\ &= (\mathbf{A} \times \mathbf{B} \cdot \mathbf{C})(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})^2 \end{aligned}$$

$$53. \text{ Given the vectors } \mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \quad \text{and} \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \text{ show that if } \mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0,$$

$$(a) \quad \mathbf{a}' \cdot \mathbf{a} = \mathbf{b}' \cdot \mathbf{b} = \mathbf{c}' \cdot \mathbf{c} = 1,$$

$$(b) \quad \mathbf{a}' \cdot \mathbf{b} = \mathbf{a}' \cdot \mathbf{c} = 0, \quad \mathbf{b}' \cdot \mathbf{a} = \mathbf{b}' \cdot \mathbf{c} = 0, \quad \mathbf{c}' \cdot \mathbf{a} = \mathbf{c}' \cdot \mathbf{b} = 0,$$

$$(c) \quad \text{if } \mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = V \text{ then } \mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}' = 1/V,$$

$$(d) \quad \mathbf{a}', \mathbf{b}', \text{ and } \mathbf{c}' \text{ are non-coplanar if } \mathbf{a}, \mathbf{b} \text{ and } \mathbf{c} \text{ are non-coplanar.}$$

$$(a) \quad \mathbf{a}' \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{a}' = \mathbf{a} \cdot \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = 1$$

$$\mathbf{b}' \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{b}' = \mathbf{b} \cdot \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{b} \cdot \mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = 1$$

$$\mathbf{c}' \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{c}' = \mathbf{c} \cdot \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{c} \cdot \mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = 1$$

$$(b) \quad \mathbf{a}' \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}' = \mathbf{b} \cdot \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{b} \cdot \mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = \frac{\mathbf{b} \times \mathbf{b} \cdot \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} = 0$$

Similarly the other results follow. The results can also be seen by noting, for example, that $\mathbf{a}'$ has the direction of $\mathbf{b} \times \mathbf{c}$ and so must be perpendicular to both $\mathbf{b}$ and $\mathbf{c}$, from which $\mathbf{a}' \cdot \mathbf{b} = 0$ and $\mathbf{a}' \cdot \mathbf{c} = 0$.

From (a) and (b) we see that the sets of vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are reciprocal vectors. See also Supplementary Problems 104 and 106.

$$(c) \quad \mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{V}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{V}, \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{V}$$

$$\begin{aligned} \text{Then } \mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}' &= \frac{(\mathbf{b} \times \mathbf{c}) \cdot (\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})}{V^3} = \frac{(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})}{V^3} \\ &= \frac{(\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})^2}{V^3} = \frac{V^2}{V^3} = \frac{1}{V} \quad \text{using Problem 52.} \end{aligned}$$

(d) By Problem 43, if $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ are non-coplanar $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \neq 0$. Then from part (c) it follows that $\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}' \neq 0$, so that $\mathbf{a}', \mathbf{b}'$ and $\mathbf{c}'$ are also non-coplanar.

54. Show that any vector $\mathbf{r}$ can be expressed in terms of the reciprocal vectors of Problem 53 as

$$\mathbf{r} = (\mathbf{r} \cdot \mathbf{a}')\mathbf{a} + (\mathbf{r} \cdot \mathbf{b}')\mathbf{b} + (\mathbf{r} \cdot \mathbf{c}')\mathbf{c}.$$

$$\text{From Problem 50, } \mathbf{B}(\mathbf{A} \cdot \mathbf{C} \times \mathbf{D}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{D}) = \mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{D}) - \mathbf{D}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})$$

$$\text{Then } \mathbf{D} = \frac{\mathbf{A}(\mathbf{B} \cdot \mathbf{C} \times \mathbf{D})}{\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}} - \frac{\mathbf{B}(\mathbf{A} \cdot \mathbf{C} \times \mathbf{D})}{\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}} + \frac{\mathbf{C}(\mathbf{A} \cdot \mathbf{B} \times \mathbf{D})}{\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}}$$

Let $\mathbf{A} = \mathbf{a}$, $\mathbf{B} = \mathbf{b}$, $\mathbf{C} = \mathbf{c}$ and $\mathbf{D} = \mathbf{r}$. Then

$$\begin{aligned} \mathbf{r} &= \frac{\mathbf{r} \cdot \mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \mathbf{a} + \frac{\mathbf{r} \cdot \mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \mathbf{b} + \frac{\mathbf{r} \cdot \mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \mathbf{c} \\ &= \mathbf{r} \cdot \left(\frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \right) \mathbf{a} + \mathbf{r} \cdot \left(\frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \right) \mathbf{b} + \mathbf{r} \cdot \left(\frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}} \right) \mathbf{c} \\ &= (\mathbf{r} \cdot \mathbf{a}')\mathbf{a} + (\mathbf{r} \cdot \mathbf{b}')\mathbf{b} + (\mathbf{r} \cdot \mathbf{c}')\mathbf{c} \end{aligned}$$

SUPPLEMENTARY PROBLEMS

55. Evaluate: (a) $\mathbf{k} \cdot (\mathbf{i} + \mathbf{j})$, (b) $(\mathbf{i} - 2\mathbf{k}) \cdot (\mathbf{j} + 3\mathbf{k})$, (c) $(2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) \cdot (3\mathbf{i} + 2\mathbf{j} - \mathbf{k})$.

Ans. (a) 0 (b) -6 (c) 1

56. If $\mathbf{A} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$, find:

(a) $\mathbf{A} \cdot \mathbf{B}$, (b) A , (c) B , (d) $|3\mathbf{A} + 2\mathbf{B}|$, (e) $(2\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - 2\mathbf{B})$.

Ans. (a) -10 (b) $\sqrt{14}$ (c) 6 (d) $\sqrt{150}$ (e) -14

57. Find the angle between: (a) $\mathbf{A} = 3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, (b) $\mathbf{C} = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\mathbf{D} = 3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}$.

Ans. (a) 90° (b) $\arccos 8/21 = 67^\circ 36'$

58. For what values of a are $\mathbf{A} = a\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{B} = 2a\mathbf{i} + a\mathbf{j} - 4\mathbf{k}$ perpendicular? Ans. $a = 2, -1$

59. Find the acute angles which the line joining the points $(1, -3, 2)$ and $(3, -5, 1)$ makes with the coordinate axes. Ans. $\arccos 2/3$, $\arccos 2/3$, $\arccos 1/3$ or $48^\circ 12'$, $48^\circ 12'$, $70^\circ 32'$

60. Find the direction cosines of the line joining the points $(3, 2, -4)$ and $(1, -1, 2)$.

Ans. $2/7, 3/7, -6/7$ or $-2/7, -3/7, 6/7$

61. Two sides of a triangle are formed by the vectors $\mathbf{A} = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. Determine the angles of the triangle. Ans. $\arccos 7/\sqrt{75}$, $\arccos \sqrt{26}/\sqrt{75}$, 90° or $36^\circ 4'$, $53^\circ 56'$, 90°

62. The diagonals of a parallelogram are given by $\mathbf{A} = 3\mathbf{i} - 4\mathbf{j} - \mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$. Show that the parallelogram is a rhombus and determine the length of its sides and its angles.

Ans. $5\sqrt{3}/2$, $\arccos 23/75$, $180^\circ - \arccos 23/75$ or 4.33 , $72^\circ 8'$, $107^\circ 52'$

63. Find the projection of the vector $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ on the vector $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. *Ans.* $8/3$
64. Find the projection of the vector $4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ on the line passing through the points $(2, 3, -1)$ and $(-2, -4, 3)$. *Ans.* 1
65. If $\mathbf{A} = 4\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{B} = -2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, find a unit vector perpendicular to both $\mathbf{A}$ and $\mathbf{B}$. *Ans.* $\pm(\mathbf{i} - 2\mathbf{j} - 2\mathbf{k})/3$
66. Find the acute angle formed by two diagonals of a cube. *Ans.* $\arccos 1/3$ or $70^\circ 32'$
67. Find a unit vector parallel to the xy plane and perpendicular to the vector $4\mathbf{i} - 3\mathbf{j} + \mathbf{k}$. *Ans.* $\pm(3\mathbf{i} + 4\mathbf{j})/5$
68. Show that $\mathbf{A} = (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})/3$, $\mathbf{B} = (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})/3$ and $\mathbf{C} = (2\mathbf{i} + \mathbf{j} - 2\mathbf{k})/3$ are mutually orthogonal unit vectors.
69. Find the work done in moving an object along a straight line from $(3, 2, -1)$ to $(2, -1, 4)$ in a force field given by $\mathbf{F} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. *Ans.* 15
70. Let $\mathbf{F}$ be a constant vector force field. Show that the work done in moving an object around any closed polygon in this force field is zero.
71. Prove that an angle inscribed in a semi-circle is a right angle.
72. Let $ABCD$ be a parallelogram. Prove that $\overline{AB}^2 + \overline{BC}^2 + \overline{CD}^2 + \overline{DA}^2 = \overline{AC}^2 + \overline{BD}^2$.
73. If $ABCD$ is any quadrilateral and P and Q are the midpoints of its diagonals, prove that $\overline{AB}^2 + \overline{BC}^2 + \overline{CD}^2 + \overline{DA}^2 = \overline{AC}^2 + \overline{BD}^2 + 4\overline{PQ}^2$
This is a generalization of the preceding problem.
74. (a) Find an equation of a plane perpendicular to a given vector $\mathbf{A}$ and distant p from the origin.
(b) Express the equation of (a) in rectangular coordinates.
Ans. (a) $\mathbf{r} \cdot \mathbf{n} = p$, where $\mathbf{n} = \mathbf{A}/A$; (b) $A_1x + A_2y + A_3z = Ap$
75. Let $\mathbf{r}_1$ and $\mathbf{r}_2$ be unit vectors in the xy plane making angles α and β with the positive x -axis.
(a) Prove that $\mathbf{r}_1 = \cos \alpha \mathbf{i} + \sin \alpha \mathbf{j}$, $\mathbf{r}_2 = \cos \beta \mathbf{i} + \sin \beta \mathbf{j}$.
(b) By considering $\mathbf{r}_1 \cdot \mathbf{r}_2$ prove the trigonometric formulas
 $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$, $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
76. Let $\mathbf{a}$ be the position vector of a given point (x_1, y_1, z_1) , and $\mathbf{r}$ the position vector of any point (x, y, z) . Describe the locus of $\mathbf{r}$ if (a) $|\mathbf{r} - \mathbf{a}| = 3$, (b) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{a} = 0$, (c) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{r} = 0$.
Ans. (a) Sphere, center at (x_1, y_1, z_1) and radius 3.
(b) Plane perpendicular to $\mathbf{a}$ and passing through its terminal point.
(c) Sphere with center at $(x_1/2, y_1/2, z_1/2)$ and radius $\frac{1}{2}\sqrt{x_1^2 + y_1^2 + z_1^2}$, or a sphere with $\mathbf{a}$ as diameter.
77. Given that $\mathbf{A} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$ are the position vectors of points P and Q respectively.
(a) Find an equation for the plane passing through Q and perpendicular to line PQ .
(b) What is the distance from the point $(-1, 1, 1)$ to the plane?
Ans. (a) $(\mathbf{r} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = 0$ or $2x + 3y + 6z = -28$; (b) 5
78. Evaluate each of the following:
(a) $2\mathbf{j} \times (3\mathbf{i} - 4\mathbf{k})$, (b) $(\mathbf{i} + 2\mathbf{j}) \times \mathbf{k}$, (c) $(2\mathbf{i} - 4\mathbf{k}) \times (\mathbf{i} + 2\mathbf{j})$, (d) $(4\mathbf{i} + \mathbf{j} - 2\mathbf{k}) \times (3\mathbf{i} + \mathbf{k})$, (e) $(2\mathbf{i} + \mathbf{j} - \mathbf{k}) \times (3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k})$.
Ans. (a) $-8\mathbf{i} - 6\mathbf{k}$, (b) $2\mathbf{i} - \mathbf{j}$, (c) $8\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$, (d) $\mathbf{i} - 10\mathbf{j} - 3\mathbf{k}$, (e) $2\mathbf{i} - 11\mathbf{j} - 7\mathbf{k}$
79. If $\mathbf{A} = 3\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$, find: (a) $|\mathbf{A} \times \mathbf{B}|$, (b) $(\mathbf{A} + 2\mathbf{B}) \times (2\mathbf{A} - \mathbf{B})$, (c) $|(\mathbf{A} + \mathbf{B}) \times (\mathbf{A} - \mathbf{B})|$.
Ans. (a) $\sqrt{195}$, (b) $-25\mathbf{i} + 35\mathbf{j} - 55\mathbf{k}$, (c) $2\sqrt{195}$
80. If $\mathbf{A} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{B} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{C} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, find:
(a) $|(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}|$, (c) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$, (e) $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{B} \times \mathbf{C})$
(b) $|\mathbf{A} \times (\mathbf{B} \times \mathbf{C})|$, (d) $(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$, (f) $(\mathbf{A} \times \mathbf{B})(\mathbf{B} \cdot \mathbf{C})$
Ans. (a) $5\sqrt{26}$, (b) $3\sqrt{10}$, (c) -20 , (d) -20 , (e) $-40\mathbf{i} - 20\mathbf{j} + 20\mathbf{k}$, (f) $35\mathbf{i} - 35\mathbf{j} + 35\mathbf{k}$
81. Show that if $\mathbf{A} \neq \mathbf{0}$ and both of the conditions (a) $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C}$ and (b) $\mathbf{A} \times \mathbf{B} = \mathbf{A} \times \mathbf{C}$ hold simultaneously then $\mathbf{B} = \mathbf{C}$, but if only one of these conditions holds then $\mathbf{B} \neq \mathbf{C}$ necessarily.
82. Find the area of a parallelogram having diagonals $\mathbf{A} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$. *Ans.* $5\sqrt{3}$

83. Find the area of a triangle with vertices at $(3, -1, 2)$, $(1, -1, -3)$ and $(4, -3, 1)$. *Ans.* $\frac{1}{2}\sqrt{165}$
84. If $\mathbf{A} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, find a vector of magnitude 5 perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.
Ans. $\pm \frac{5\sqrt{3}}{3}(\mathbf{i} + \mathbf{j} + \mathbf{k})$
85. Use Problem 75 to derive the formulas
 $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$, $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
86. A force given by $\mathbf{F} = 3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$ is applied at the point $(1, -1, 2)$. Find the moment of $\mathbf{F}$ about the point $(2, -1, 3)$. *Ans.* $2\mathbf{i} - 7\mathbf{j} - 2\mathbf{k}$
87. The angular velocity of a rotating rigid body about an axis of rotation is given by $\boldsymbol{\omega} = 4\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. Find the linear velocity of a point P on the body whose position vector relative to a point on the axis of rotation is $2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$. *Ans.* $-5\mathbf{i} - 8\mathbf{j} - 14\mathbf{k}$
88. Simplify $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{B} + \mathbf{C}) \times (\mathbf{C} + \mathbf{A})$. *Ans.* $2\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$
89. Prove that $(\mathbf{A} \cdot \mathbf{B} \times \mathbf{C})(\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}) = \begin{vmatrix} \mathbf{A} \cdot \mathbf{a} & \mathbf{A} \cdot \mathbf{b} & \mathbf{A} \cdot \mathbf{c} \\ \mathbf{B} \cdot \mathbf{a} & \mathbf{B} \cdot \mathbf{b} & \mathbf{B} \cdot \mathbf{c} \\ \mathbf{C} \cdot \mathbf{a} & \mathbf{C} \cdot \mathbf{b} & \mathbf{C} \cdot \mathbf{c} \end{vmatrix}$
90. Find the volume of the parallelepiped whose edges are represented by $\mathbf{A} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$, $\mathbf{B} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{C} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. *Ans.* 7
91. If $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = 0$, show that either (a) $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are coplanar but no two of them are collinear, or (b) two of the vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are collinear, or (c) all of the vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are collinear.
92. Find the constant a such that the vectors $2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $3\mathbf{i} + a\mathbf{j} + 5\mathbf{k}$ are coplanar. *Ans.* $a = -4$
93. If $\mathbf{A} = x_1\mathbf{a} + y_1\mathbf{b} + z_1\mathbf{c}$, $\mathbf{B} = x_2\mathbf{a} + y_2\mathbf{b} + z_2\mathbf{c}$ and $\mathbf{C} = x_3\mathbf{a} + y_3\mathbf{b} + z_3\mathbf{c}$, prove that

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})$$
94. Prove that a necessary and sufficient condition that $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ is $(\mathbf{A} \times \mathbf{C}) \times \mathbf{B} = \mathbf{0}$. Discuss the cases where $\mathbf{A} \cdot \mathbf{B} = 0$ or $\mathbf{B} \cdot \mathbf{C} = 0$.
95. Let points P , Q and R have position vectors $\mathbf{r}_1 = 3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$, $\mathbf{r}_2 = \mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ and $\mathbf{r}_3 = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ relative to an origin O . Find the distance from P to the plane OQR . *Ans.* 3
96. Find the shortest distance from $(6, -4, 4)$ to the line joining $(2, 1, 2)$ and $(3, -1, 4)$. *Ans.* 3
97. Given points $P(2, 1, 3)$, $Q(1, 2, 1)$, $R(-1, -2, -2)$ and $S(1, -4, 0)$, find the shortest distance between lines PQ and RS . *Ans.* $3\sqrt{2}$
98. Prove that the perpendiculars from the vertices of a triangle to the opposite sides (extended if necessary) meet in a point (the *orthocenter* of the triangle).
99. Prove that the perpendicular bisectors of the sides of a triangle meet in a point (the *circumcenter* of the triangle).
100. Prove that $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) + (\mathbf{B} \times \mathbf{C}) \cdot (\mathbf{A} \times \mathbf{D}) + (\mathbf{C} \times \mathbf{A}) \cdot (\mathbf{B} \times \mathbf{D}) = 0$.
101. Let PQR be a spherical triangle whose sides p, q, r are arcs of great circles. Prove the *law of cosines for spherical triangles*,

$$\cos p = \cos q \cos r + \sin q \sin r \cos P$$

with analogous formulas for $\cos q$ and $\cos r$ obtained by cyclic permutation of the letters.

[Hint: Interpret both sides of the identity $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{C}) = (\mathbf{B} \cdot \mathbf{C})(\mathbf{A} \cdot \mathbf{A}) - (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{A})$.]

102. Find a set of vectors reciprocal to the set $2\mathbf{i}+3\mathbf{j}-\mathbf{k}$, $\mathbf{i}-\mathbf{j}-2\mathbf{k}$, $-\mathbf{i}+2\mathbf{j}+2\mathbf{k}$.

$$\text{Ans. } \frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{k}, \quad -\frac{8}{3}\mathbf{i} + \mathbf{j} - \frac{7}{3}\mathbf{k}, \quad -\frac{7}{3}\mathbf{i} + \mathbf{j} - \frac{5}{3}\mathbf{k}$$

103. If $\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$, $\mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$ and $\mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$, prove that

$$\mathbf{a} = \frac{\mathbf{b}' \times \mathbf{c}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}, \quad \mathbf{b} = \frac{\mathbf{c}' \times \mathbf{a}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}, \quad \mathbf{c} = \frac{\mathbf{a}' \times \mathbf{b}'}{\mathbf{a}' \cdot \mathbf{b}' \times \mathbf{c}'}$$

104. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{a}', \mathbf{b}', \mathbf{c}'$ are such that

$$\mathbf{a}' \cdot \mathbf{a} = \mathbf{b}' \cdot \mathbf{b} = \mathbf{c}' \cdot \mathbf{c} = 1$$

$$\mathbf{a}' \cdot \mathbf{b} = \mathbf{a}' \cdot \mathbf{c} = \mathbf{b}' \cdot \mathbf{a} = \mathbf{b}' \cdot \mathbf{c} = \mathbf{c}' \cdot \mathbf{a} = \mathbf{c}' \cdot \mathbf{b} = 0$$

prove that it necessarily follows that

$$\mathbf{a}' = \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{b}' = \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}, \quad \mathbf{c}' = \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}}$$

105. Prove that the only right-handed self-reciprocal sets of vectors are the unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

106. Prove that there is one and only one set of vectors reciprocal to a given set of non-coplanar vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$.